Infrared Spectroscopy of Atmospheric Trace Gases(Greenhouse Effect)

Advanced Lab course
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1 Introduction

The Earth's radiative balance is strongly influenced by the absorption of infrared (IR) radiation by atmospheric trace gases. While the atmosphere is largely transparent to incoming solar radiation in the visible range, several molecular species absorb outgoing terrestrial radiation in the infrared spectral region. This absorption process forms the physical basis of the greenhouse effect.

Molecules interact with electromagnetic radiation when the radiation frequency matches allowed transitions between quantized molecular energy levels.

In the infrared region, these transitions correspond primarily to vibrational and rotational motions of molecules. Only molecules whose vibration leads to a change in electric dipole moment are IR active.

Greenhouse gases such as CO_2, H_2O, and CH_4 possess vibrational modes that produce dipole moment changes and therefore absorb infrared radiation efficiently. In contrast, homonuclear diatomic molecules such as N_2 and O_2 do not exhibit dipole moment changes during vibration and are therefore largely IR inactive.

In this experiment, infrared absorption spectra of atmospheric gases were measured using a Fourier Transform Infrared (FTIR) spectrometer. The aim was to identify characteristic absorption bands of greenhouse gases and relate these spectral features to molecular structure and the physical mechanism of the greenhouse effect.

2 Theoretical Background

2.1 Infrared Region

The infrared region spans approximately

$$\lambda \approx 0.7\,\mu\text{m to } 1000\,\mu\text{m}. \tag{1}$$

In spectroscopy, the wavenumber $\tilde{\nu}$ is commonly used:

$$\tilde{\nu} = \frac{1}{\lambda}. \tag{2}$$

Molecular vibrational transitions typically occur in the mid-infrared region:

$$4000\text{–}400\,\text{cm}^{-1}. \tag{3}$$

2.2 Molecular Degrees of Freedom

For a molecule consisting of N atoms:

$$\text{Total degrees of freedom} = 3N \tag{4}$$
$$\text{Translational degrees} = 3 \tag{5}$$

Rotational degrees of freedom:

- Linear molecule: 2
- Nonlinear molecule: 3

Vibrational degrees of freedom:

- Linear molecule: $3N - 5$
- Nonlinear molecule: $3N - 6$

Examples:
CO_2 (linear, $N = 3$):

$$3N - 5 = 4 \text{ vibrational modes} \tag{6}$$

H_2O (nonlinear, $N = 3$):

$$3N - 6 = 3 \text{ vibrational modes} \tag{7}$$

2.3 Vibrational Energy of a Diatomic Molecule

In the harmonic oscillator approximation:

$$E_v = \hbar\omega \left(v + \frac{1}{2} \right), \tag{8}$$

where $v = 0, 1, 2, \dots$
Infrared transitions mainly follow the selection rule:

$$\Delta v = \pm 1. \tag{9}$$

2.4 Rotational Energy of a Diatomic Molecule

For a rigid rotor:

$$E_J = BJ(J + 1), \tag{10}$$

with

$$B = \frac{\hbar^2}{2I}, \tag{11}$$

where $J = 0, 1, 2, \dots$ and I is the moment of inertia.
Selection rule:

$$\Delta J = \pm 1. \tag{12}$$

The combination of vibrational and rotational transitions leads to rovibrational bands consisting of P- and R-branches.

2.5 Rovibrational Transitions

To obtain a spectrum from rotational and vibrational energy levels, selection rules must be determined that specify which transitions are allowed. These rules follow from the transition dipole moment matrix elements

$$|\langle J'M'|\hat{\mu}_i|JM\rangle|^2 = \left| \int \psi^*_{J'M'}\, \hat{\mu}_i\, \psi_{JM}\, d^3r \right|^2 . \tag{13}$$

A transition is allowed only if this integral is nonzero. Whether it vanishes or not is determined by the symmetry properties of the molecular wavefunctions.

For a diatomic molecule in the harmonic oscillator and rigid rotor approximation, the selection rules are:

Vibrational

$$\Delta v = \pm 1 \tag{14}$$

Rotational

$$\Delta J = \pm 1 \tag{15}$$

Since vibrational energy spacings are much larger than rotational energy spacings ($E_{\mathrm{vib}} \gg E_{\mathrm{rot}}$), infrared radiation excites vibrational and rotational degrees of freedom simultaneously. Pure vibrational transitions without rotational contribution are therefore not observed.

Energy Levels

Restricting the consideration to transitions from the vibrational ground state ($v = 0$) to the first excited state ($v = 1$), the rovibrational energy levels are given by

$$E(v, J) = h\nu_0 \left(v + \frac{1}{2} \right) + BJ(J+1), \tag{16}$$

where ν_0 is the fundamental vibrational frequency and

$$B = \frac{\hbar^2}{2I} \tag{17}$$

is the rotational constant.

Transition Energies

The energy difference for a rovibrational transition becomes

$$\Delta E = h\nu_0 \pm 2BJ. \tag{18}$$

Thus, transitions with $\Delta J = +1$ or $\Delta J = -1$ produce lines with positive or negative frequency shifts relative to the pure vibrational frequency ν_0.

Band Structure

The observed spectrum, known as a rovibrational band, is divided into:

- P-branch ($\Delta J = -1$)
- R-branch ($\Delta J = +1$)

Because transitions with $\Delta J = 0$ are forbidden, no spectral line appears exactly at ν_0. The spacing between neighboring lines is

$$\nu(J+1) - \nu(J) = 2B, \tag{19}$$

which shows that all lines within a branch are equally spaced.

2.6 Intensity Distribution and J_{max}

The intensity distribution within a rovibrational absorption band of a diatomic molecule is determined by the thermal population of rotational states and their degeneracy.

At thermal equilibrium, the population of a rotational level J follows Boltzmann statistics:

$$N_J \propto (2J+1) \exp\left(-\frac{BJ(J+1)}{k_B T}\right). \tag{20}$$

The factor $(2J+1)$ accounts for the rotational degeneracy due to the possible orientations of the angular momentum.

For small J, the degeneracy term dominates and the population increases with J. For larger J, the exponential Boltzmann factor becomes dominant and the population decreases again.

Therefore, the line intensities within the P- and R-branches increase up to a maximum and then decrease.

To determine the position of the intensity maximum, J_{max}, J is treated as a continuous variable and the derivative of the population function is set to zero:

$$\frac{d}{dJ}\left[(2J+1)\exp\left(-\frac{BJ(J+1)}{k_B T}\right)\right] = 0. \tag{21}$$

This leads to the approximate expression

$$J_{\text{max}} \approx \sqrt{\frac{k_B T}{2B}} - \frac{1}{2}. \tag{22}$$

Hence, J_{max} increases with temperature and decreases with increasing rotational constant B.

2.7 Fourier-Transform Spectrometer

A Fourier-Transform Spectrometer (FTIR) is used to measure infrared absorption spectra with high spectral resolution and high signal-to-noise ratio. Instead of dispersing light into individual frequencies, the FTIR records an interferogram in the optical path difference domain and converts it into a spectrum by Fourier transformation.

Principle of Operation

The central component of an FTIR spectrometer is a Michelson interferometer consisting of:

- Infrared source
- Beam splitter
- Fixed mirror
- Movable mirror
- Detector

The beam splitter divides the incoming broadband infrared radiation into two beams. One beam is reflected toward a fixed mirror, while the other is transmitted toward a movable mirror. After reflection, the two beams return to the beam splitter, where they recombine and interfere with each other.

The movable mirror changes position during the measurement, which varies the optical path difference between the two beams in a controlled manner. This variation produces constructive and destructive interference depending on the wavelength and path difference. As a result, the detected intensity changes as a function of mirror displacement, generating an interference signal called an interferogram. This interferogram contains information about all wavelengths present in the source simultaneously. By applying a Fourier transform to the interferogram, the spectral intensity of the infrared radiation can be obtained as a function of wavenumber.

For a given wavelength, the detected intensity varies with mirror position due to constructive and destructive interference. The positions of maximum and minimum intensity depend on the wavelength, since the phase difference between the interfering beams is determined by the optical path difference.

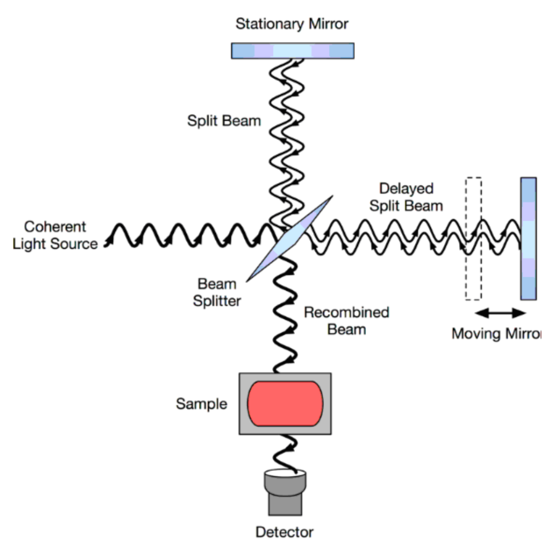


Figure 1: Schematic setup of a Michelson interferometer used in an FTIR spectrometer.

Because the interferometer does not separate wavelengths, the detector measures only the total intensity, which is the sum of the contributions from all wavelengths present. Thus, the measured signal represents the superposition of all spectral components.

If the light source emits n discrete wavelengths, measuring the intensity at n different mirror positions provides a system of n equations that can be solved to determine the spectral intensities. Increasing the maximum mirror displacement improves the spectral resolution and allows closely spaced wavelengths to be resolved.

2.8 Lambert–Beer Absorption Law

According to the Lambert–Beer absorption law, the transmitted intensity at frequency ν after passing through an absorbing medium is given by,

$$I(\nu) = I_0(\nu)\, \exp\left[-k(\nu)\, p\, l\right], \tag{23}$$

where

- $I_0(\nu)$ is the incident intensity,
- $I(\nu)$ is the transmitted intensity,
- $k(\nu)$ is the absorption coefficient $[\mathrm{cm^{-1}atm^{-1}}]$,
- p is the gas pressure [atm],
- l is the absorption path length [cm].

Column Density Form

Alternatively, the Lambert–Beer law can be expressed in terms of the column density N:

$$I(\nu) = I_0(\nu)\, \exp\left[-\beta(\nu)\, N\right], \tag{24}$$

where

- $\beta(\nu)$ is the molecular absorption cross section $[\mathrm{cm^2}]$,
- N is the column density, i.e. the total number of absorbing particles per unit area $[\mathrm{cm^{-2}}]$.

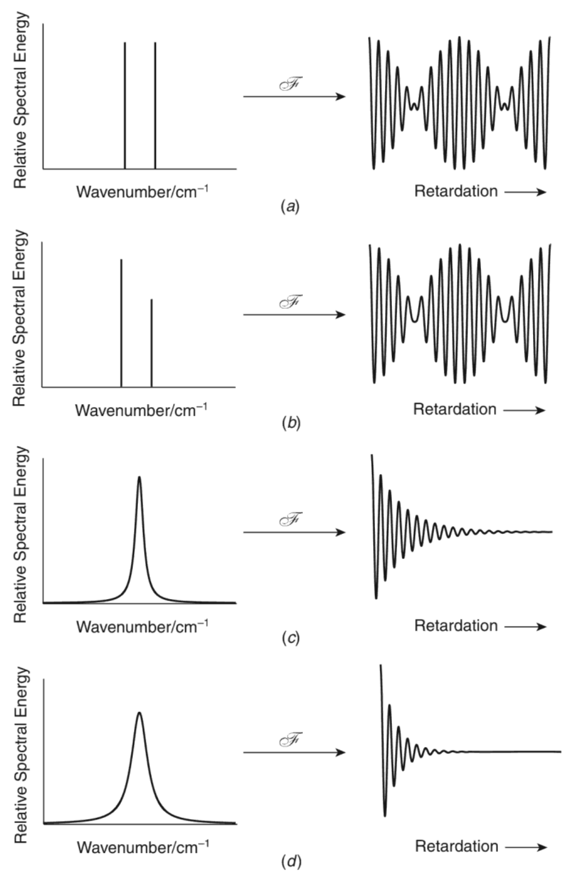


Figure 2: Examples of spectra (left) and the corresponding interferograms (right) [3]

Absorption Line Strength

The strength of an absorption line or an entire absorption band is obtained by integrating the absorption coefficient over frequency:

$$S = \int k(\nu)\, d\nu. \tag{25}$$

The integrated line strength S is an intrinsic molecular property and is independent of pressure and path length.

2.9 Barometric Altitude Formula

The barometric altitude formula describes the decrease of atmospheric pressure with increasing altitude under the assumption of hydrostatic equilibrium and constant temperature.

The pressure at height h is given by

$$p(h) = p(h_0)\, \exp\left(-\frac{gM(h - h_0)}{RT}\right), \tag{26}$$

where

- $p(h)$ is the pressure at altitude h,
- $p(h_0)$ is the pressure at reference altitude h_0 (e.g. sea level),
- $g = 9.81\,\mathrm{m\,s^{-2}}$ is the gravitational acceleration,
- M is the molar mass of the gas,
- $R = 8.314\,\mathrm{J\,mol^{-1}\,K^{-1}}$ is the universal gas constant,
- T is the absolute temperature,
- $(h - h_0) = \Delta h$ is the altitude difference.

Particle Density

Using the ideal gas law,

$$p = n k_B T, \tag{27}$$

the particle number density $n(h)$ becomes

$$n(h) = \frac{p(h)}{k_B T(h)}. \tag{28}$$

The total number of particles along the atmospheric column is obtained by integrating the particle density over height:

$$N = \int_0^{h_{\max}} n(h)\, dh. \tag{29}$$

Temperature Profile in the Troposphere

In the troposphere (up to approximately 10 km), the temperature decreases approximately linearly with altitude. The average lapse rate is

$$\Gamma \approx 6\,\mathrm{K\,km^{-1}}. \tag{30}$$

The temperature at altitude h can therefore be approximated by

$$T(h) = T(h_0) - \Gamma(h - h_0), \tag{31}$$

where

- $T(h)$ is the temperature at altitude h,
- $T(h_0)$ is the temperature at sea level,
- Γ is the temperature lapse rate.

Above the troposphere, the particle density decreases significantly and the temperature profile deviates from this linear approximation.

3 Experimental Methods

3.1 Apparatus

The measurements were performed using a Fourier Transform Infrared (FTIR) spectrometer consisting of:

- Broadband infrared source
- Michelson interferometer
- Beam splitter
- Fixed mirror
- Movable mirror
- Gas cell / atmospheric path
- Infrared detector
- Computer for Fourier transformation

3.2 Measurement Principle

The Michelson interferometer splits the incident radiation into two beams. One beam is reflected by a fixed mirror, the other by a movable mirror. The recombined beams interfere depending on the optical path difference. The recorded interferogram is Fourier transformed to obtain the spectrum.

3.3 Procedure

1. The spectrometer was aligned and initialized.
2. A background spectrum was recorded.
3. Atmospheric air was measured under laboratory conditions.
4. Spectra were recorded in the mid-infrared region.
5. Interferograms were Fourier transformed by the instrument software.
6. Absorbance spectra were obtained by comparison with the background measurement.

4 Evaluation of the measured spectra

After performing the experiment, we obtained the spectra of N_2O and laboratory air. Due to unclear weather conditions, we were not able to obtain the spectrum of the Sun. However, the required solar spectrum was later provided to us. The following sections evaluate the spectra.

4.1 Description of the spectra

The absorption spectrum of the Sun is typical for the infrared (IR) region, with distinct absorption bands of CO_2 and H_2O. In the spectrum of laboratory air, we also observe absorption bands of CO_2 and H_2O. However, they are much less pronounced because the path length of the light through the atmosphere is much larger than the path length of the light through laboratory air.

The measured solar spectrum has a maximum at the wavenumber of 2615.67 cm^{-1}. This is not consistent with the expected maximum of blackbody radiation. This is because we are measuring the spectrum in the infrared region, which lies away from the peak of the solar radiation spectrum, which is in visible range (20000 cm^{-1}). The second reason is the Earth's atmosphere. The atmosphere absorbs a significant amount of radiation and therefore shifts the observed maximum. This further complicates the determination of the true maximum of the solar spectrum.

In addition, the sensitivity of the instrument varies for different wavenumbers, which also contributes to the shift of the maximum.

The spectrum is also dense with absorption bands. This is caused by typical IR-absorbing molecules. This can be seen in figure 3.

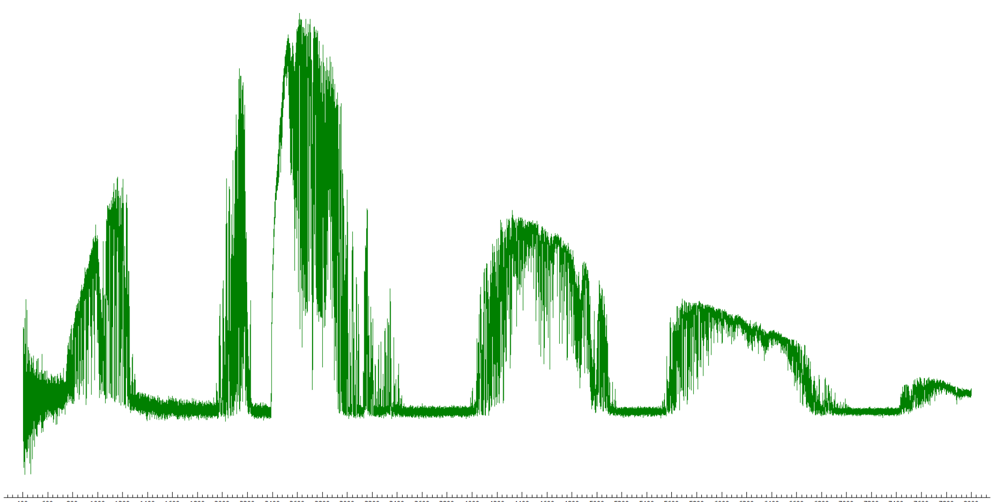


Figure 3: Absorption spectrum of the Sun with dense regions of absorption bands.

The solar spectrum is very wide compared to the other two, spanning from 300 cm^{-1} to 8000 cm^{-1}. This allows only limited comparison with the other spectra. We managed to identify the three most prominent absorption bands. In figure 4, we have identified the corresponding molecules responsible for these absorption bands. The most prominent molecule is water. Because of its V-shaped structure, it has many vibrational modes, which are given by $3N - 6$ for non-linear molecules. Due to its three fundamental vibrations, H_2O does not show clear P and R branches compared to CO_2, which is the second molecule identified. The visible P and R branches are a result of the linear structure of CO_2.

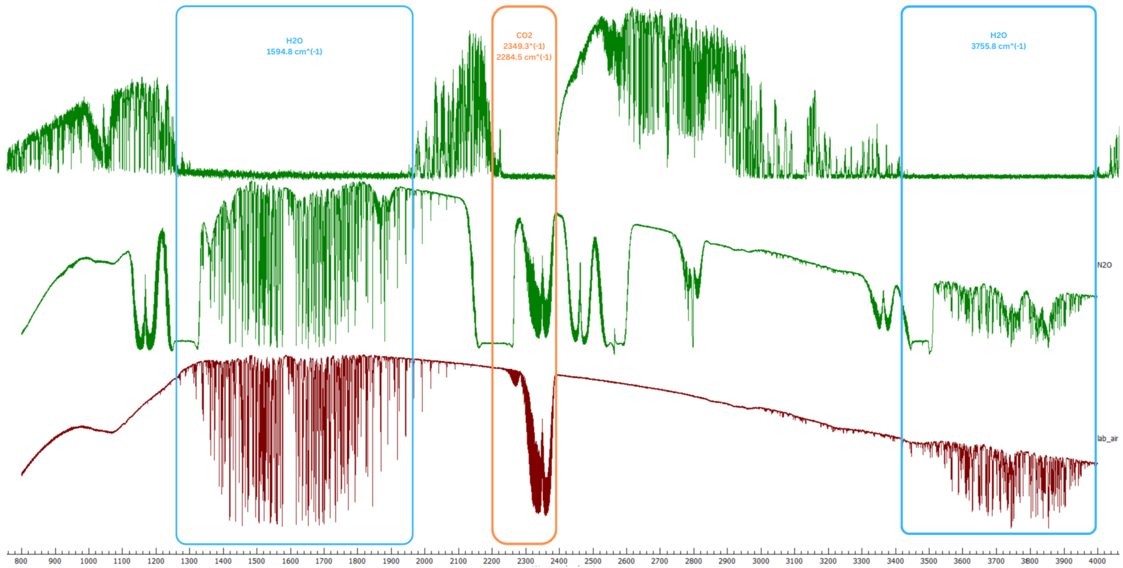


Figure 4: Comparison of the spectra of the Sun (top), N_2O, and laboratory air, showing absorption bands.

4.2 Blackbody Temperature

By analyzing the intensity maximum of the solar spectrum, we can calculate the effective temperature of the Sun using Wien's Displacement Law:

$$\lambda_{\text{max}} = \frac{b}{T} \tag{32}$$

where $b \approx 0.29$ cm $\cdot$ K.

Given the maximum of the spectrum at 2615.67 cm^{-1}, the calculated temperature of the Sun was found to be 758.54 K. This value is significantly lower than the actual solar effective temperature of approximately 5772 K. This discrepancy is primarily due to atmospheric extinction and the specific spectral characteristics discussed in Section 4.1.

Additionally, we calculated the temperatures of the laboratory air and N_2O using the same methodology. The resulting temperatures were 527.69 K and 415.2 K, respectively. These results are also lower than the actual temperature of the laser source used in the experiment (approximately 1000 K), which is likely attributable to the limited sensitivity and calibration of the instrument.

4.3 Identification of Molecules

A detailed examination of the solar spectra revealed numerous absorption bands exhibiting distinct P and R branches. These results are summarized in Table 1.

Observed Band Origin (cm^{-1})	**Literature Value** (cm^{-1})	**Identified Molecule**
2462.6	2461.5	N_2O
2563.3	2563.5	N_2O
4855.5	4860.5	CO_2
4977.8	4983.5	CO_2
6228.4	6231.0	CO_2
6345.8	6351.0	CO_2

Table 1: Observed and literature band origins of identified atmospheric trace gases.

Apart from the expected species, we observed a prominent absorption band near 1044.3 cm^{-1}. This band featured well-defined P and R branches, as shown in Figure 5. While the identity of this molecule is not immediately certain, the most likely candidate for a strong absorption band in this region is ozone (O_3). Although ozone is a non-linear molecule and typically exhibits more complex rotational structures, the literature identifies a major ozone band at 1042 cm^{-1} [2]. This aligns closely with our experimental observation, though the simplicity of the observed branches warrants further investigation.

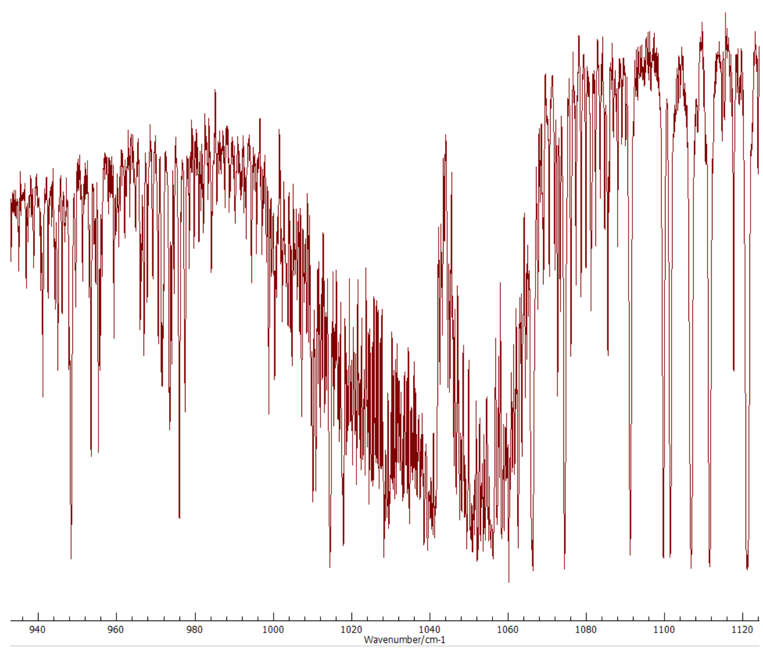


Figure 5: Observed absorption band near 1044.3 cm^{-1} displaying P and R branches.

4.4 Calculation of Moment of Inertia

Molecular spectra provide significant insights into the physical properties of a molecule. Using the N_2O spectrum, we can calculate the molecular moment of inertia (I) via its relationship with the rotational constant, B, as defined in Equation 17.

To perform this calculation, we selected an absorption band with well-resolved P and R branches in the region of 1168.8 cm^{-1} (see Figure 6). By measuring the distance between five consecutive rotational lines, we determined the average line spacing, which corresponds to $2B$.

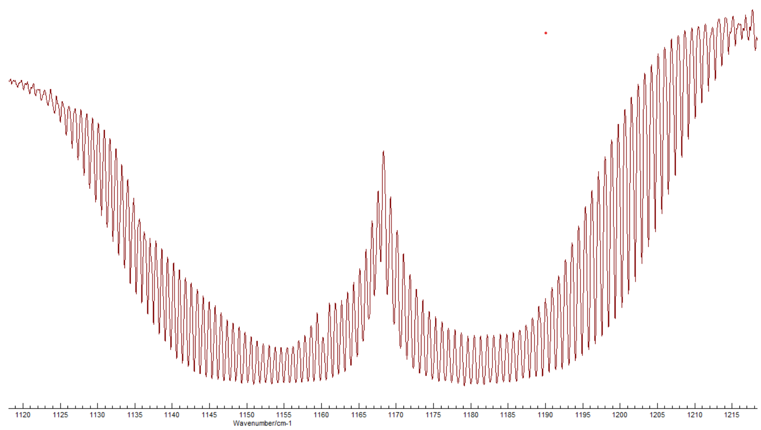


Figure 6: N_2O absorption band selected for the moment of inertia calculation.

The moment of inertia for N_2O derived from the spectra was 6.689×10^{-47} kg $\cdot$ m^2. Compared to the literature value of 6.680×10^{-47} kg $\cdot$ m^2, our experimental result shows an error of only 0.13%.

4.5 Determination of Gas Pressure

The gas pressure within the cell was determined using the Beer-Lambert Law, the theoretical basis of which was established in Section 2.8. By utilizing the absorption line strength (S), the pressure can be calculated using the following relation:

$$\ln\left(\frac{I}{I_0}\right) = -S \cdot p \cdot L \cdot \frac{1}{\Delta\nu} \tag{33}$$

where $L = 15$ cm is the path length of the cell, p is the partial pressure of the gas, and S is the line strength (taken as 10.9 cm$^{-2} \cdot$ atm^{-1} for the specific transition).

In this analysis, I represents the intensity of the absorption line, while I_0 is the baseline intensity (or the integrated area under the curve for the reference). After performing the relevant calculations, the pressure of the gas sample inside the cell was found to be 0.4161 atm.

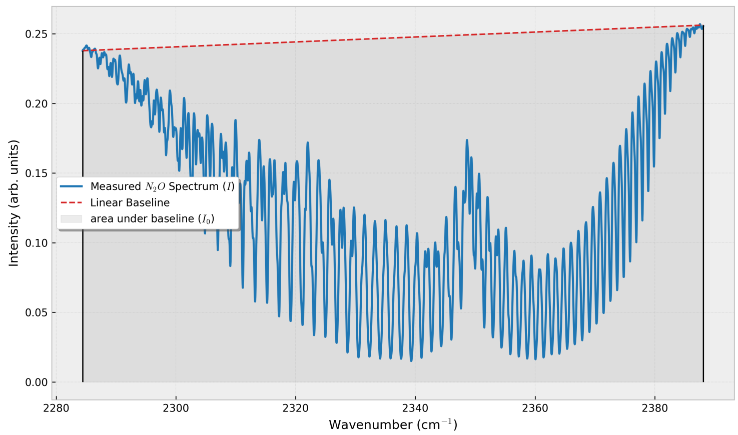


Figure 7: N_2O absorption band selected for gas pressure calculation through integration.

4.6 Temperature Determination of N_2O

To determine the rotational temperature of the gas sample, we employed the population distribution relationship discussed in Section 2.6 (Equation 22).

This method requires identifying $J_{\max}$, the rotational quantum number corresponding to the maximum absorption intensity within a branch. We determined $J_{\max}$ from two distinct absorption bands to ensure consistency.

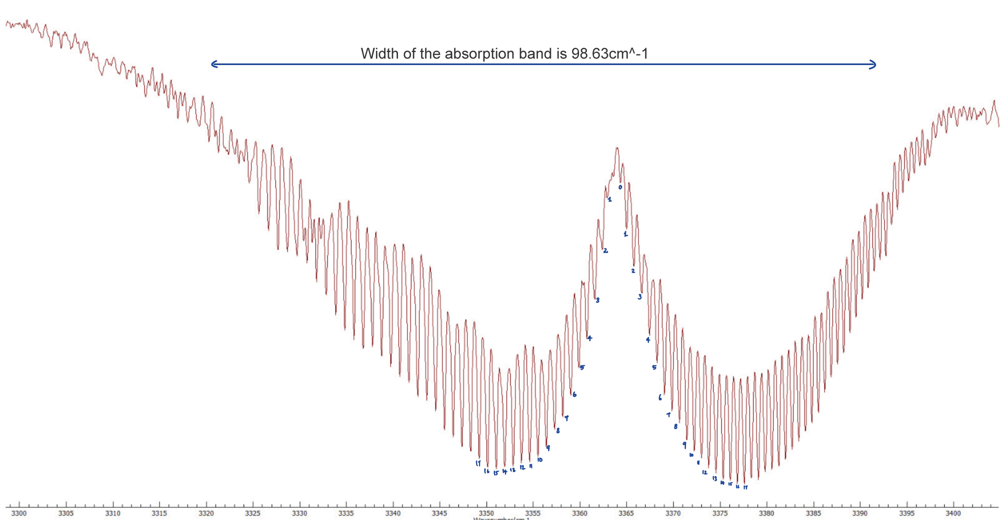


Figure 8: N_2O absorption band used to identify $J_{\max}$ for temperature calculation.

Using the determined $J_{\max}$ values, the gas temperature was calculated for both instances. The first band yielded a temperature of 289.769 K, while the second band resulted in a temperature of 328.364 K.

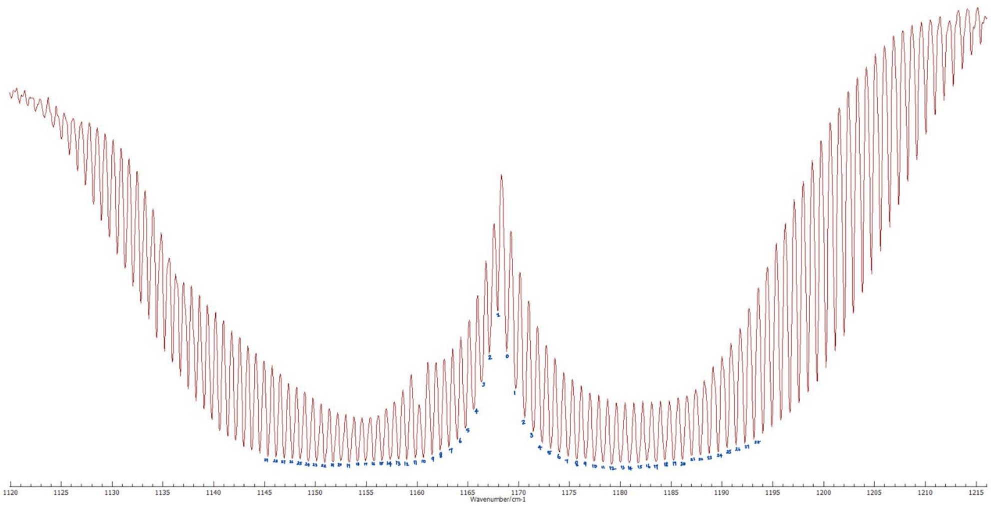


Figure 9: Secondary N_2O absorption band used for $J_{\max}$ verification.

4.7 Determination of Greenhouse Gas Concentrations

To determine the concentration of specific gases in the atmosphere, it is first necessary to calculate the total number of molecules in the atmospheric column. This is achieved by applying the Ideal Gas Law (Equation 28). By using the pressure and temperature at sea level as a reference, we can estimate the particle density at the surface and subsequently extrapolate this value to higher altitudes using the Barometric Formula (Equation 26).

Under these conditions, the particle density scales with altitude as illustrated in Figure 10.

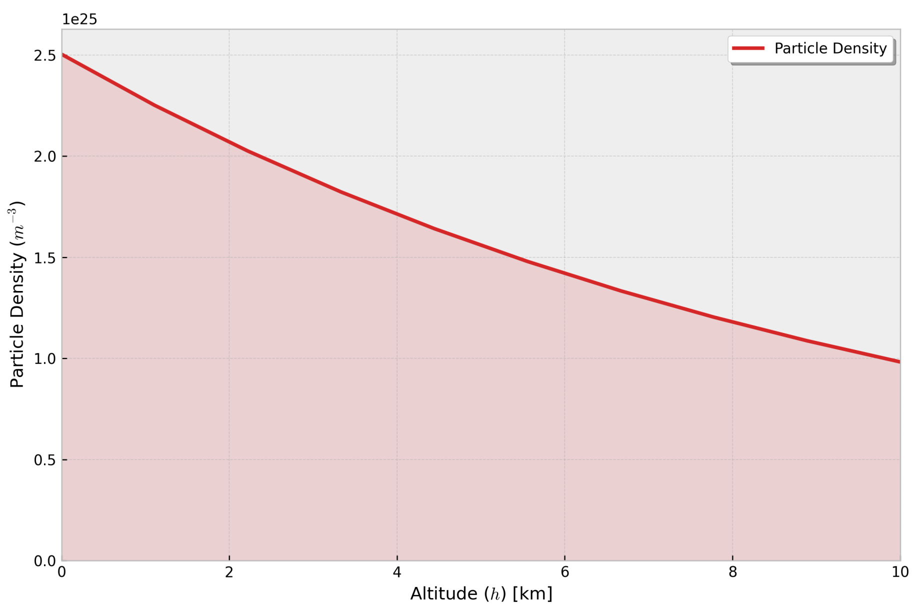


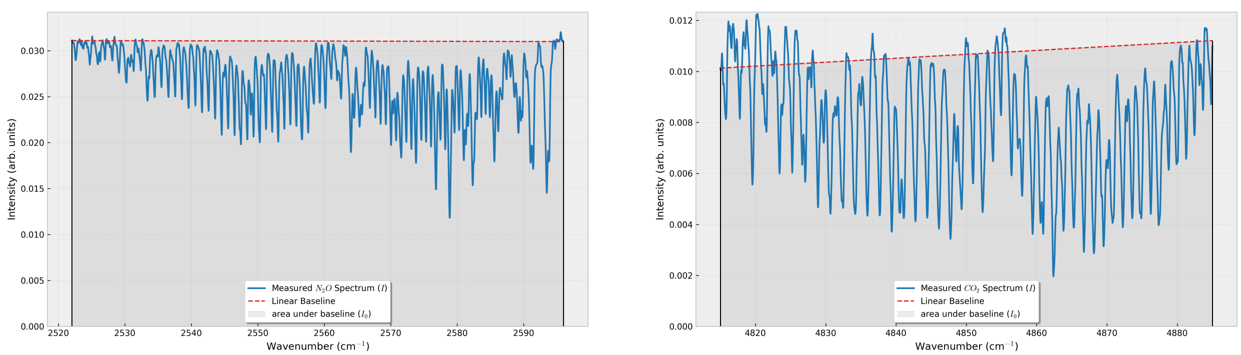
Figure 10: Modeled particle density as a function of altitude.

By integrating the particle density over the atmospheric height, we obtain the total column density, N_0. Our calculations yielded a value of $N_0 = 1.619 \times 10^{25}$ cm^{-2}. To account for the total number of particles encountered by the solar radiation, the optical path length must be adjusted based on the solar zenith angle. For the spectra provided, the angle of inclination (α) was 59°. The total number of irradiated particles, N_{ges}, is calculated as follows:

$$N_{\text{ges}} = \frac{N_0}{\sin\alpha} = 2.5428 \times 10^{25} \text{ cm}^{-2} \tag{34}$$

With the total atmospheric particle count established, we can estimate the concentration of specific trace gases using the Beer-Lambert Law in its column density form (Equation 24).

To determine the incident intensity (I_0), we applied a linear baseline integration, as illustrated in Figures 11a and 11b for N_2O and CO_2, respectively.



(a) N_2O band (2522–2596 cm^{-1})

(b) CO_2 band (4815–4885 cm^{-1})

Figure 11: Integration of absorption bands for N_2O and CO_2 to determine atmospheric particle numbers.

The absorption coefficients (β) for these bands were sourced from literature [1]. Using these coefficients, the number of molecules in the atmosphere was determined to be 1.34×10^{19} cm^{-2} for CO_2 and 3.01×10^{21} cm^{-2} for N_2O.

The resulting volume mixing ratios in PPM (parts per million) and PPB (parts per billion) are compared with standard literature values in Table 2.

Source	N_2O **[PPB]**	CO_2 **[PPM]**
Calculated	1028.88	116
Literature	≈ 325	≈ 400

Table 2: Comparison between calculated concentrations and literature values.

Significant deviations were observed for both N_2O and CO_2. These discrepancies likely stem from spectral noise and the presence of overlapping absorption features. Noise in the baseline can artificially "lift" the spectrum, resulting in unrealistic intensity ratios and significant calculation errors.

Furthermore, we noted that the edges of the CO_2 spectrum do not return cleanly to the baseline, with portions of the signal appearing above the integrated reference line. Because the Beer-Lambert Law depends logarithmically on the intensity ratio, even minor baseline offsets can lead to drastic changes in the final concentration. The errors are then can be the result of our simplified model of atmosphere.

4.8 Summary

In this experiment, the infrared absorption spectra of the sun, laboratory air, and a contained N_2O gas sample were analyzed to investigate molecular constants and atmospheric trace gas concentrations.

The solar spectrum revealed a complex structure dominated by H_2O and CO_2 absorption bands. The measured intensity maximum at 2615.67 cm^{-1} yielded a brightness temperature of 758.54 K via Wien's displacement law. This value is significantly lower than the actual solar surface temperature due to the selective sensitivity of the detector in the IR region and the heavy attenuation of shorter wavelengths by the Earth's atmosphere.

Key molecular parameters were successfully extracted from the N_2O spectra:

- **Moment of Inertia:** Calculated as 6.689×10^{-47} kg$\cdot$ m^2, which showed an excellent agreement with the literature value (6.680×10^{-47} kg$\cdot$ m^2), resulting in a relative error of only 0.13%.
- **Gas Pressure:** Determined via the Lambert-Beer law to be 0.4161 atm within the sample cell.
- **Rotational Temperature:** Analysis of J_{max} in two different bands provided temperatures of 289.77 K and 328.36 K, reflecting the thermal state of the gas sample during measurement.

Atmospheric concentrations for N_2O and CO_2 were estimated using column density integration. While the calculated values (1028.88 ppb and 116 ppm respectively) showed noticeable deviations from standard atmospheric averages, these were attributed to the difficulty in defining a precise baseline (I_0) due to spectral noise and overlapping absorption features.

References

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[3] Peter R. Griffiths and James A. de Haseth. *Fourier Transform Infrared Spectrometry*. Wiley-Ineterscience, 2007.